§7 Indeterminate Forms and L'Hospital's Rule

* <u>L'Hospital's Rule</u>

$f,\ g$: differentiable functions

$g'(x) \neq 0$ on an open interval I that contains a (except possibly at a)

$\left(\lim_{x\to a} f(x)=0 \text{ and } \lim_{x\to a} g(x)=0\right)$ or $\left(\lim_{x\to a} f(x)=\pm\infty \text{ and } \lim_{x\to a} g(x)=\pm\infty\right)$

$\Rightarrow \lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$

if the limit on the right side exists (or is ∞ or $-\infty$)

(proof of the special case)

* "$x\to a$" can be replaced by any of the symbols $x\to a^+$, $x\to a^-$, $x\to\infty$ and $x\to-\infty$.

Exam 1 Find $\lim_{x\to 1}\frac{\ln x}{x-1}$.

(sol)

Exam 2 Calculate $\lim_{x\to\infty}\frac{e^x}{x^2}$.

(sol)

Exam 3 Calculate $\lim_{x\to\infty}\frac{\ln x}{\sqrt[3]{x}}$.

(sol)

Exam 4 Find $\lim_{x\to 0}\frac{\tan x - x}{x^3}$.

(sol)

Exam 5 Find $\displaystyle\lim_{x\to\pi^-}\frac{\sin x}{1-\cos x}$.

(sol)

$*$ **Indeterminate Products (** $0\cdot\infty$ **)**

$$f\cdot g=\frac{g}{1/f}\quad\text{or}\quad f\cdot g=\frac{f}{1/g}$$

$$\left(0\cdot\infty\ \Rightarrow\frac{\infty}{\infty}\quad\text{or}\quad 0\cdot\infty\ \Rightarrow\frac{0}{0}\right)$$

Exam 6 Evaluate $\displaystyle\lim_{x\to0^+}x\ln x$.

(sol)

∗ **Indeterminate Differences ($\infty - \infty$)**

$$f - g = \frac{\frac{1}{g} - \frac{1}{f}}{\frac{1}{fg}} \quad \left(\infty - \infty \ \Rightarrow \ \frac{0}{0} \right)$$

Exam 7 Compute $\lim_{x \to (\pi/2)^-} (\sec x - \tan x)$.

(sol)

* **Indeterminate Powers** (0^0, ∞^0, 1^∞)

Let $y = \{f(x)\}^{g(x)}$.

$\Rightarrow \ln y = g(x) \ln f(x)$ ($\Rightarrow 0 \cdot (-\infty)$, $0 \cdot \infty$, $\infty \cdot 0$)

If $\lim_{x \to a} \ln y = \lim_{x \to a} g(x) \ln f(x) = \alpha$, then

$$\lim_{x \to a} \{f(x)\}^{g(x)} = \lim_{x \to a} y = \lim_{x \to a} e^{\ln y} = e^{\alpha}.$$

Exam 8 Calculate $\lim_{x \to 0^+} (1 + \sin 4x)^{\cot x}$.

(sol)

Exam 9 Find $\lim_{x \to 0^+} x^x$.

(sol)

(Exam)

1. Find $\lim_{x \to 0^+} \frac{e^{-1/x}}{x}$.

(sol)

2. Find $\lim_{x \to \infty} (x - \ln x)$

(sol)

3. $\lim_{x\to 0}\frac{\sin^{-1}x}{x}=$

4. $\lim_{x\to 0}\frac{\sinh^{-1}x}{x}=$

5. $\lim_{x\to 0}\frac{\cosh x-1}{x^2}=$

6. Evaluate $\lim_{x\to 0^+}(\sinh^{-1}x)^x$

(sol)